

Kimberly Ruth

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Education

Stanford University , Stanford, CA Ph.D. Computer Science, 2026 M.S. Computer Science, 2023 Advised by Prof. Zakir Durumeric Thesis: <i>Rebuilding Empirical Foundations for Security Research and Decision-Making</i>	2020 - 2026
University of Washington , Seattle, WA B.S. in Computer Engineering, B.A. in Mathematics College Honors; Tau Beta Pi; Phi Beta Kappa; <i>summa cum laude</i> GPA 3.99 Co-advised by Profs. Franziska Roesner and Tadayoshi Kohno Thesis: <i>Understanding and Designing for Security and Privacy in Multi-User AR Interactions</i>	2015 - 2020

Positions

Postdoctoral Fellow, University of Chicago Data Science Institute , Chicago, IL <i>Faculty mentor: Prof. Grant Ho</i>	8/2026 - Present
Engineering Practicum Intern, Google , Mountain View, CA <i>Team: Security and privacy engineering</i>	6/2016 - 9/2016

Selected Awards and Honors

*National recognition marked with **

University of Chicago Data Science Institute Postdoctoral Fellowship	2026
* IRTF Applied Networking Research Prize	2026
* NSF Graduate Research Fellowship (GRFP)	2020
Stanford EDGE Fellowship	2020
* CRA Outstanding Undergraduate Researcher Award Winner	2020
UW Allen School Best Senior Thesis Award	2020
UW Allen School Outstanding Senior	2020
UW College of Engineering Dean's Medalist	2020
* CRA Outstanding Undergraduate Researcher Award Finalist	2018, 2019
* Barry Goldwater Scholar	2018
UW Husky 100 Award	2018
Washington Research Foundation Fellowship	2017, 2018, 2019
* ACSA Scholarship for Women Studying Information Security	2017, 2018
University of Washington Mary Gates Research Scholar	2017

Publications

Peer-Reviewed

Conference

- [1] **K. Ruth**, R. B. Obu, I. Shode, G. Li, C. Gates, G. Ho, and Z. Durumeric, "A first look at governments' enterprise security guidance," in *Proceedings of the 34th USENIX Security Symposium*, USENIX Association, August 2025
- [2] **K. Ruth**, V. A. Rivera, G. Akiwate, A. Fass, P. G. Kelley, K. Thomas, and Z. Durumeric, "'Perfect is the enemy of good': The CISO's role in enterprise security as a business enabler," in *ACM CHI Conference on Human Factors in Computing Systems (CHI'25)*, ACM, April 2025
- [3] R. Habib, **K. Ruth**, G. Akiwate, and Z. Durumeric, "Formalizing dependence of web infrastructure," in *Proceedings of the ACM SIGCOMM 2025 Conference*, ACM, September 2025
IRTF Applied Networking Research Prize
- [4] **K. Ruth**, A. Fass, J. J. Azose, M. Pearson, E. Thomas, C. Sadowski, and Z. Durumeric, "A world wide view of browsing the world wide web," in *ACM Internet Measurement Conference (IMC'22)*, ACM, October 2022
- [5] **K. Ruth**, D. Kumar, B. Wang, L. Valenta, and Z. Durumeric, "Toppling top lists: Evaluating the accuracy of popular website lists," in *ACM Internet Measurement Conference (IMC'22)*, ACM, October 2022
- [6] M. Ziv, L. Izhikevich, **K. Ruth**, K. Izhikevich, and Z. Durumeric, "ASdb: A system for classifying owners of Autonomous Systems," in *ACM Internet Measurement Conference (IMC'21)*, ACM, November 2021
- [7] **K. Ruth**, T. Kohno, and F. Roesner, "Secure multi-user content sharing for augmented reality applications," in *Proceedings of the 28th USENIX Security Symposium*, USENIX Association, August 2019
- [8] K. Lebeck, **K. Ruth**, T. Kohno, and F. Roesner, "Towards security and privacy for multi-user augmented reality: Foundations with end users," in *Proceedings of the 39th IEEE Symposium on Security and Privacy (Oakland)*, IEEE, May 2018
- [9] K. Lebeck, **K. Ruth**, T. Kohno, and F. Roesner, "Securing augmented reality output," in *Proceedings of the 38th IEEE Symposium on Security and Privacy (Oakland)*, IEEE, May 2017

Invited Article

- [10] K. Lebeck, **K. Ruth**, T. Kohno, and F. Roesner, "Arya: Operating system support for securely augmenting reality," *IEEE Security and Privacy Magazine*, vol. 16, pp. 44–53, February 2018

Non-Refereed

- [11] C. Patrick, **K. Ruth**, and Z. Durumeric, "The code the world depends on: A first look at technology makers' open source software dependencies," 2024
- [12] *Contributed to*: University of Washington Security and Privacy Research Lab and UW Reality Lab, "2019 industry-academia summit on mixed reality security, privacy, and safety: Summit report," April 2020. Ed. Franziska Roesner and Tadayoshi Kohno.

Teaching Experience

CS 356 Course Assistant, Stanford University, Stanford, CA 9/2025-12/2025

- Served as TA for 30-person “Topics in Computer and Network Security” graduate security course
- Ran 2 class sessions: moderated student presentations and paper discussions
- Contributed to updating paper reading list
- Graded weekly paper reading responses; assisted with grading project deliverables

CS 249i Course Assistant, Stanford University, Stanford, CA 1/2025-3/2025

- Served as TA for 36-person “The Modern Internet” upper-division networking course
- Gave 80-minute guest lecture on security, eCrime, and attacks; held weekly office hours
- Assisted with reworking assignments and rubrics and grading projects and final exam

CS 249i Course Assistant, Stanford University, Stanford, CA 1/2024-3/2024

- Served as sole student-facing TA for 65-person “The Modern Internet” upper-division networking course
- Gave 80-minute guest lecture on security, eCrime, and attacks; held weekly office hours
- Assisted with provisioning infrastructure and grading projects and final exam

CS 356 Course Assistant, Stanford University, Stanford, CA 9/2023-12/2023

- Served as TA for 36-person “Topics in Computer and Network Security” graduate security course
- Ran 5 class sessions (25% of the course): moderated student presentations and paper discussions
- Contributed to updating paper reading list
- Graded weekly paper reading responses; assisted with grading project deliverables

TA/Grader, Art of Problem Solving, San Diego, CA 4/2015 - 12/2019

- Guided 50+ highly motivated middle and high school students as teacher assistant in fast-paced online introductory and intermediate-level Python programming classes
- Assisted online math classes as substitute TA as needed (from pre-algebra through pre-calculus)
- Wrote detailed suggestions for Intermediate Python course content based on observations as TA
- Named AoPS Python All-Star for consistently high quality of Python grader written feedback

Peer Tutor, University of Washington Allen School, Seattle, WA 10/2017 - 6/2018

- Served as volunteer tutor for CSE 311 (Foundations of Computing), an introduction to mathematics and proofs for computer science and engineering majors
- Ran weekly tutoring meetings for 3 students; prepared original practice materials beforehand
- Communicated status to tutoring program organizers for further development of undergraduate tutoring program

Computer Science Tutor, Independent Consultant, Bellevue, WA 9/2015 - 1/2016

- Worked with student on introductory-level Python programming
- Taught code design, development, debugging, and documentation

Advising and Mentoring

Undergraduate and Master's Student Researchers

Karena Gau (Summer 2025), Stanford BS '28 (expected)
 Divya Ganesan (Academic Year 2024-25), Stanford BA Political Science / MS CS '26 → Palantir
 Tara Dixit (Academic Year 2024-25), Stanford BS CS '27 (expected)
 Vedant Singh (Winter 2025), Stanford BS CS '26 → Salus (YC W26)
 Raymond Buernor Obu (Academic Year 2024-26), Stanford MS CS '26 → Amazon
 Gavin Li (Academic Year 2024-25), Stanford MS CS '25 → Apple
 Ifeoluwa Shode (Summer and Academic Year 2024-25), Fisk University BS CS '25 (expected)
 Anna Ascherman (Spring 2024), Stanford BS/MS CS '24 → CISA
 Cadence Patrick (Summer 2023), Spelman BS CS '25 → Northwestern MS '26 (expected)
 Modupe Akintan (Academic Year 2022-24), Stanford MS CS '24 → Amazon
 Sasha Ronaghi (Academic Year 2022-23), Stanford BS/MS CS '25
 Aditya Saligrama (Academic Year 2022-25), Stanford BS/MS CS '24 → Formal
 America Sophia George (Summer 2022), Stanford BS Economics '23
 Basheerah Abdus-Shakur (Summer 2022), University of the District of Columbia BS CS '23
 → Lockheed Martin
 Nathan Bhak (Summer 2021), Stanford BS/MS CS '24 → Palo Alto Networks
 → Johns Hopkins Applied Physics Lab
 Luca Pistor (Summer 2021), Stanford BS/MS CS '23 → Citadel
 Maya Ziv (Academic Year 2020-21), Stanford BS/MS CS '21 → Censys → Pavilion
 Henry Bowman (Summer 2019), Cal Poly San Luis Obispo BS CS '19 → Moonbeam → Amazon
 AJ Kruse (Summer 2019); UW BS CS

Other Mentorship

Mentor, Stanford EDGE Fellowship Program, 9/2022-6/2024
 Mentor, Stanford CS Mentor Program, 10/2020-6/2024
 Mentor, UW SWEsters, 11/2016-6/2019

Professional Service

External

Program Committees

IEEE S&P 2027
 CCS 2026
 RAID 2024
 RAID 2023 – Especially Noteworthy Reviewer
 SOUPS Poster Jury 2023
 SOUPS Poster Jury 2022

Invited Reviews

IEEE S&P (Oakland) subreview, 2024
 USENIX subreview, 2024
 IEEE S&P Magazine invited review, 2023
 EuroS&P subreview, 2023
 IEEE S&P (Oakland) subreview, 2023
 USENIX subreview, 2022
 IMC subreview, 2021
 NSDI subreview, 2021

Department

Stanford-Berkeley Women in CS/EE Research Meetup organizer

Lead organizer, 1/2024-4/2024

- Lead organization of all-day event hosted at Stanford for women and nonbinary researchers at all career levels
- Coordinate 8 volunteers across 2 campuses to arrange talks, panels, a poster session, and networking time
- Event attracted over 100 RSVPs

Supporting lead, 3/2025-4/2025

- Managed announcements and carpool coordination on Stanford side of Berkeley-hosted event
- Event attracted over 40 Stanford RSVPs

Ph.D. Student Admissions Committee member, 12/2022 - 2/2023

Student-Applicant Support Program reviewer, 11/2022, 11/2023, 11/2024, 11/2025

Stanford Security Reading Group organizer, 7/2022 - 12/2025

Stanford Security Lunch organizer, 7/2021 - 8/2022

- Ran weekly lunch seminar attracting 25-40 attendees in hybrid meeting format
- Invited and coordinated with both internal and external presenters

Women's Research Day undergraduate liaison, 12/2017 - 4/2018, 1/2019 - 4/2019, 1/2020 - 4/2020

- Represented undergraduate perspective on organizing committee for UW Allen School Women's Research Day event
- Recruited 5 female undergraduate researchers each year as poster presenters and lightning talk presenters
- Presented on current research via lightning talk in 2018 and via poster in 2019 and 2020

Volunteering

Panelist, CS Virtual Preview Day, 10/2023

Panelist, EDGE Perspectives on Identifying Your Research Project Panel, 2/2023

External reviewer, Stanford Intersect (undergraduate-run journal), 5/2022

Panelist, CURIS Graduate School Panel, 8/2021, 9/2021
 Poster judge, Stanford Undergraduate Research Conference, 4/2021
 Panelist and Mentor, Stanford/Berkeley Graduate Pathways to STEM, 2/2021-5/2021
 Panelist, Student Advisory Council Research Panel, 5/2020
 Speaker and Panelist, ACM Research Night, 2/2020
 Volunteer, UW Math Circle, 10/2019
 Lightning talk speaker, Kobe University Scholars visit, 8/2019
 Lightning talk speaker, Undergraduate Research Lightning Talks at UW Libraries, 5/2019
 Panelist, Admitted Student Preview Day Research Panel, 4/2019
 Panelist, Direct Admit Freshman Seminar, 10/2016, 11/2017, 10/2018
 Lightning talk speaker, CSE Undergraduate Research Lunch, 8/2018
 Volunteer, UW Engineering Discovery Days, 4/2018
 Mentor, ACM New Student Welcome, 9/2016, 3/2017, 9/2017
 Panelist and Mentor, Direct Admit "CSE Startup" course, 8/2017
 Panelist, SWE Undergraduate Research Panel, 11/2016

Talks and Posters

Talks

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| [T1] | A First Look at Governments' Enterprise Security Guidance
USENIX Security 2025, Seattle, WA | August 2025 |
| [T2] | Rethinking Support for CISOs' Security Decisions
UCSD Security Lunch, San Diego, CA
UC Santa Cruz SRL Seminar, Santa Cruz, CA
Stanford Computer Forum Security Workshop, Stanford, CA | June 2025
May 2025
April 2025 |
| [T3] | "Perfect is the Enemy of Good": The CISO's Role in Enterprise Security as a Business Enabler
ACM CHI Conference on Human Factors in Computing Systems, Yokohama, Japan | April 2025 |
| [T4] | A World Wide View of the World Wide Web
UW Security Seminar, Seattle, WA
MIT Security Seminar, Virtual
CISPA Web Security Seminar, Virtual
Stanford Security Lunch, Stanford, CA
UCSD Security Lunch, San Diego, CA
Berkeley Security Seminar, Berkeley, CA | Feb 2023
Feb 2023
Jan 2023
Dec 2022
Oct 2022
Oct 2022 |
| [T5] | A World Wide View of Browsing the World Wide Web
Internet Measurement Conference (IMC), Nice, France | Oct 2022 |
| [T6] | Toppling Top Lists: Evaluating the Accuracy of Popular Website Lists
Internet Measurement Conference (IMC), Nice, France | Oct 2022 |
| [T7] | Understanding and Designing for Security and Privacy in Multi-User AR Interactions
Stanford Security Lunch, Stanford, CA
Undergraduate Research Symposium, Seattle, WA
Microsoft HoloLens Team Brown Bag Lunch Seminar, Seattle, WA | Sept 2020
May 2020
March 2020 |
| [T8] | Understanding and Designing for Security and Privacy in Multi-User AR Interactions
Industry-Academia Summit on Mixed Reality Security, Privacy, and Safety, Seattle, WA | Sept 2019 |

- [T9] **Secure Multi-User Content Sharing for Augmented Reality Applications**
USENIX Security 2019, Santa Clara, CA Aug 2019
- [T10] **Secure Multi-User Content Sharing for Augmented Reality Applications**
Microsoft Research Crypto Day, Redmond, WA June 2019
Undergraduate Research Symposium, Seattle, WA May 2019
- [T11] **Designing for Security and Privacy in Multi-User Augmented Reality Interactions**
Undergraduate Research Symposium, Seattle, WA May 2018
- [T12] **Emerging Security and Privacy Challenges in Augmented Reality**
UW ACM Student Tech Talk, Seattle, WA April 2018
- [T13] **Securing Application Output of Augmented Reality Systems**
Undergraduate Research Symposium, Seattle, WA May 2017

Posters

- [P1] **Secure Multi-User Content Sharing for Augmented Reality Applications**
UW Allen School Women's Research Day 2020, Seattle, WA April 2020
- [P2] **Secure Multi-User Content Sharing for Augmented Reality Applications**
UW Allen School Affiliates Research Day 2019, Seattle, WA Nov 2019
- [P3] **Secure Multi-User Content Sharing for Augmented Reality Applications**
UW Allen School Women's Research Day, Seattle, WA April 2019